



CSE623 Machine Learning: Theory and Practice

Topic:

Machine Learning Based Risk Parity Portfolio Construction for Financial Markets

Weekly Report 4

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1. Overview of Project:

Risk parity has emerged as a widely adopted portfolio construction framework that allocates capital by equalizing risk contributions across assets rather than maximizing expected returns. Popularized in institutional asset management as an all weather strategy, risk parity portfolios are designed to deliver stable performance across different market regimes. However, the effectiveness of risk parity critically depends on accurate estimation of asset volatilities and correlations.

Recent advances in machine learning and statistical learning theory provide tools for denoising and dynamically estimating covariance structures. Techniques such as shrinkage estimators, random matrix theory based eigenvalue filtering, and hierarchical clustering provide more robust representations of dependency structures tail risks for a multi-asset portfolio.

This project builds a risk parity portfolio using machine learning techniques. Using daily US financial assets data from 2015–2025, we construct long-short, net-long, fully invested portfolios. We then evaluate their out-of-sample performance from the year 2020 till 2025 and compare it with a benchmark.

2. Overview of Week 4:

Tasks accomplished this week:

- The full PCA Risk Parity pipeline was refactored and simplified into a modular, production-ready codebase. Each stage of the pipeline, data preparation, covariance estimation, factor selection, regularization, weight computation, and back-testing—was implemented as an independent function, allowing systematic experimentation with different configurations.
- Ridge regularization was introduced on the PCA-reduced covariance matrix to address the extreme weight magnitudes that were reaching $\pm 9,900\%$ reported in Week 3. An eigenvalue floor and an additive ridge penalty were applied before inverting the covariance, stabilizing the resulting portfolio weights.
- A weight explosion guard was implemented as the rebalancing threshold mechanism. If newly computed weights deviate beyond a multiplicative threshold relative to the previous period's weights, the portfolio retains the prior allocation rather than accepting the unstable new weights.
- A rolling back-test was executed with a four-year training window and 21-day monthly holding period using Marčenko–Pastur denoised covariance. Out-of-sample performance metrics were computed including total return, annualized return, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, and Calmar ratio.
- The S&P 500 index was used as the benchmark. A rolling 252-day correlation between the strategy returns and the S&P 500 returns was computed to assess how the portfolio's risk exposures co-move with the broad equity market over time.

3. Work Completed:

Pipeline Refactoring and Code Simplification

- The full pipeline was refactored from its earlier multi-method experimental form into a clean, modular architecture. Each function encapsulates a single responsibility: `prepare_data()` handles loading, yield-to-price conversion, log-return computation, and missing value removal; `raw_cov()`,

ledoit_wolf_cc(), and marchenko_pastur_denoise() implement the three covariance estimators; select_factors() performs eigen decomposition with optional detoning; regularized_cov() applies ridge regularization; and compute_weights() derives the closed-form minimum variance weights.

- The earlier multi-asset dataset included U.S. government bond yields (USGG3YR and USGG10YR). Since covariance estimation requires price-based return series rather than yield levels, a yield-to-price conversion function was implemented. This converts each yield observation into a synthetic zero-coupon bond price using the formula $P = FV / (1 + y)^T$, where FV is the face value (1,000), y is the annualized yield, and T is the maturity in years (3 and 10, respectively). Log returns were then computed on these synthetic prices, bringing bond instruments onto a consistent return basis with equities, commodities, and currencies.

Detoning: Removal of the Market Factor

- An optional detoning step was implemented in the factor selection stage. When enabled, the first principal component which typically captures the broad market factor is removed before selecting the remaining signal components. The rationale is that PC1 often dominates the covariance structure and concentrates portfolio risk in a single systematic direction. By removing it, the subsequent risk parity allocation operates over genuinely diversifying latent factors (sector rotation, cross-asset volatility, yield curve movements) rather than being dominated by the equity market mode.
- The select_factors() function performs eigen decomposition, sorts eigenvalues in descending order, optionally drops the first component, and then dynamically determines how many remaining components to retain. The selection criterion retains all principal components whose eigenvalue exceeds the mean eigenvalue, which adapts the number of factors to the signal structure present in each rolling window.

Ridge Regularization of the Reduced Covariance Matrix

- A key issue identified in Week 3 was that PCA-reduced covariance matrices are often near-singular, causing the matrix inverse used in weight computation to amplify estimation noise into extreme portfolio weights. To address this, a two-part regularization procedure was introduced.
- First, an eigenvalue floor is applied: any eigenvalue below a minimum threshold (set to 10^{-3}) is replaced with that threshold. This prevents near-zero eigenvalues from producing arbitrarily large entries in the inverse. Second, a ridge penalty ($\lambda = 10^{-3}$) is added to every eigenvalue before reconstructing the regularized covariance as $\Sigma_{\text{reg}} = V_k \text{diag}(\lambda_1 + \lambda, \lambda_2 + \lambda, \dots, \lambda_k + \lambda) V_k^T$. The combined effect is a well-conditioned covariance matrix whose inverse yields stable, practically feasible portfolio weights.

Closed-Form Minimum Variance Weights on PCA-Reduced Covariance

- Portfolio weights were computed using the closed-form global minimum variance solution applied to the ridge-regularized PCA-reduced covariance matrix. The formula is:

$$w = \Sigma^{-1} \mathbf{1} / (\mathbf{1}^T \Sigma^{-1} \mathbf{1})$$

- where Σ is the regularized covariance matrix and $\mathbf{1}$ is the vector of ones. This solution minimizes portfolio variance subject to the constraint that weights sum to one. Because the covariance is

constructed in the reduced PCA space and then regularized, the resulting weights inherit the denoised, diversified factor structure while remaining numerically stable.

- Individual asset weights may be positive (long) or negative (short), but the portfolio is net-long and fully invested ($\sum w_i = 1$). This long-short structure is consistent with the framework discussed in Roncalli and Weisang 2012 and resolves the negative-return issue encountered in Week 3 when a strict long-only constraint was imposed on inherently signed factor exposures.

Weight Explosion Guard (Rebalancing Threshold)

- To prevent sudden, destabilizing changes in portfolio allocation, a weight explosion guard was implemented. At each rebalancing date, the newly computed weight vector is compared against the previous period's weights. If the minimum weight falls below 1.25 times the previous minimum, or the maximum weight exceeds 1.25 times the previous maximum, the new weights are rejected and the portfolio retains its prior allocation.
- This mechanism serves as the rebalancing threshold discussed in the Week 3 plan. Rather than calibrating a fixed distance metric, the guard uses a multiplicative deviation check on the extreme weights, which directly addresses the weight explosion phenomenon observed in earlier experiments. The threshold factor of 1.25 was selected through empirical experimentation to balance responsiveness to genuine allocation shifts against robustness to transient estimation noise.

Rolling Back-Test Execution

- The rolling back-test was executed using a four-year training window ($252 \times 4 = 1,008$ trading days) and a 21-trading-day holding period, corresponding to approximately monthly rebalancing. The Marčenko–Pastur denoised covariance estimator was used for the primary run, with a ridge regularization parameter of $\lambda = 10^{-3}$.
- At each rebalancing date, the pipeline selects the trailing four years of log returns, estimates the covariance matrix, performs PCA with optional detoning, applies ridge regularization, computes minimum variance weights, passes them through the explosion guard, and then evaluates portfolio performance over the subsequent 21-day forward window. The number of assets with non-negligible weight $|w_i| > 10^{-6}$ is tracked at each rebalance and annotated on the equity curve for diagnostic purposes.

Performance Evaluation Framework

- A comprehensive performance evaluation function was implemented computing the following metrics from the equity curve: total return, annualized return from log returns scaled by 252 trading days, annualized volatility, Sharpe ratio by assuming zero risk-free rate, Sortino ratio by using downside deviation only, information ratio, maximum drawdown.
- The preliminary results from the Marčenko–Pastur configuration are summarized below:

Table 1: Out-of-Sample Performance Metrics (Marčenko–Pastur + Ridge Regularization)

Metric	Value
Total Return	3.91%
Annualized Return	0.55%
Annualized Volatility	29.33%

Sharpe Ratio	0.02
Sortino Ratio	0.03
Information Ratio	0.02
Maximum Drawdown	−57.75%
Calmar Ratio	0.01

- The results indicate that while the regularization and guard mechanisms have successfully eliminated the extreme weight explosions reported in Week 3, the strategy’s risk-adjusted performance remains weak. The annualized volatility of 29.33% is high relative to the 0.55% annualized return, producing a near-zero Sharpe ratio. The maximum drawdown of −57.75% suggests that the portfolio experienced severe losses during adverse market regimes, indicating that the current factor selection and regularization parameters require further tuning.

Benchmark Comparison: S&P 500 Rolling Correlation

- The S&P 500 total return index was adopted as the benchmark. Daily log returns of the S&P 500 were computed from price data obtained via Bloomberg. The strategy and benchmark return series were aligned to the same date range.
- A 252-day one-year rolling correlation between the strategy returns and the S&P 500 returns was computed and plotted. This rolling correlation diagnostic reveals how the portfolio’s factor exposures co-move with the broad equity market over time. Periods of high positive correlation indicate that the portfolio is behaving like a leveraged equity bet, while periods of low or negative correlation suggest genuine diversification across non-equity risk factors. This analysis will be used in subsequent weeks to evaluate whether detoning and alternative covariance methods improve diversification relative to the equity benchmark.

4. Challenges Encountered:

- Although ridge regularization eliminated the most extreme weight magnitudes previously reaching $\pm 9,900\%$, the resulting portfolio still exhibits high annualized volatility 29.33% and a severe maximum drawdown −57.75%. The regularization parameter λ and the eigenvalue floor interact in complex ways with the number of retained principal components, and finding an optimal joint configuration requires systematic grid search or cross-validation, which has not yet been completed.
- The choice of training window length presents a bias-variance trade-off. The current four-year window 1,008 days provides a large sample for covariance estimation but may include stale regimes that no longer reflect current market dynamics. A shorter window would adapt faster but would increase estimation noise, particularly for a multi-asset universe of this size. Experimentation with window lengths ranging from two to five years is needed.
- The weight explosion guard, while effective at preventing catastrophic weight jumps, introduces path dependence into the back-test. When the guard rejects a new weight vector, the portfolio carries forward stale allocations that may no longer reflect the current covariance structure. If rejections persist across several consecutive rebalancing dates, the portfolio can drift significantly

from the intended risk parity profile. A more sophisticated approach such as a graduated blending between old and new weights may be needed.

- Preliminary results show near-zero risk-adjusted returns despite the theoretical appeal of PCA risk parity. This suggests that the closed-form minimum variance approach on the PCA-reduced covariance may not be the ideal objective for this asset universe, and alternative allocation schemes such as Hierarchical Risk Parity or direct Equal Risk Contribution optimization should be explored.

5. Plan for Week 5:

- Implement Hierarchical Risk Parity as an alternative allocation method. HRP uses hierarchical clustering on the correlation matrix to construct a tree-based allocation that does not require matrix inversion, potentially avoiding the instability issues encountered with the minimum variance approach.
- Run the rolling back-test across all three covariance estimators Ledoit–Wolf, Marčenko–Pastur and compare their out-of-sample performance head-to-head using the evaluation framework built this week.
- Build a blended benchmark that matches the asset class composition of the portfolio which is equities, bonds, commodities, currencies, REITs to provide a fairer performance comparison than the equity-only S&P 500.

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